

NATIONAL ECONOMICS UNIVERSITY
NEU – COLLEGE OF TECHNOLOGY



REPORT OF INTRODUCTION TO DATA SCIENCE

TOPIC: MARKETING DATA ANALYSIS IN BANKING

GROUP 5: NGUYỄN THỊ NGỌC KHÁNH
 CHU KHÁNH LY
 ĐẶNG NGUYỄN THẢO MY
Class: DS67B

Supervising lecture: Dr. Lương Văn Thiện

Hanoi, December 2025

Declaration

We hereby declare that all the contents contained in this project entitled “MARKETING DATA ANALYSIS IN BANKING” are the result of our own independent work and research effort. All analyses, discussions, data collection, and conclusions in this report were carried out and completed by the members of our group, without any references from other authors.

We firmly affirm that this project has not been submitted for evaluation in any other program or for any academic purpose or qualification at the National Economics University.

The work was conducted independently at the National Economics University, with all collaborations clearly stated in the text and Acknowledgements.

Group 5
November 2025

Acknowledgements

We would like to express our sincere gratitude to all individuals and institutions whose support contributed to the successful completion of this project.

First, we are deeply thankful to our supervisor, Dr. Luong Van Thien, for their expert guidance, constructive feedback, and continuous encouragement throughout the development of this study. Their insights were fundamental in shaping our research direction and ensuring the quality of our work.

We would also like to acknowledge the Faculty of Mathematical Economics at the National Economics University, particularly the Data Science in Economics and Business program for providing essential resources, academic support, and a positive learning environment. Their dedication played an important role in facilitating our progress.

Our appreciation also extends to our classmates and peers, whose valuable discussions, collaboration, and shared knowledge greatly assisted us in overcoming challenges and refining our methodology.

Lastly, we would like to express our heartfelt gratitude to our families for their unwavering support, patience, and encouragement. Their belief in our abilities was a constant source of motivation during the entire project.

Abstract

This project takes a closer look at the Bank Marketing dataset with the aim of predicting whether a customer is likely to subscribe to a term deposit after a marketing campaign. The dataset includes 41,188 customer records, covering a wide range of information such as demographic profiles, personal and financial characteristics, previous interactions with the bank, and relevant macroeconomic conditions.

To approach the problem systematically, the study follows a complete data analysis workflow. This includes conducting exploratory data analysis (EDA) to understand the structure and behavior of the data, performing preprocessing steps to clean and prepare the dataset, encoding categorical features, and finally building and evaluating machine learning models. Throughout this process, the analysis focuses on identifying meaningful patterns, understanding variable distributions, and ensuring the dataset is properly split into training and testing sets for fair evaluation.

Several machine learning models were tested, and the LightGBM consistently delivered the strongest performance in terms of accuracy and overall stability. The findings reveal that certain features, particularly call duration, key economic indicators like euribor3m and emp.var.rate, and customers past contact history, play an especially important role in predicting the final outcome.

Overall, the results show how machine learning can be a powerful tool for banks in understanding customer behaviors and improving the efficiency of marketing strategies. Beyond offering insights into the dataset, the study provides a practical foundation for developing predictive systems that can support real-world decision-making in the banking sector.

Table of Contents

- List of Figuresix
- List of Tablesx
- 1. INTRODUCTION1
 - 1.1. Background1
 - 1.2. Problem Statement1
 - 1.3. Research Objectives1
 - 1.4. Scope of Study1
 - 1.5. Organization of the Report.....1
- 2. LITERATURE REVIEW2
 - 2.1. Machine Learning for Classification Problems2
 - 2.2. Synthesis of Relevant Datasets2
 - 2.3. Challenges in Predicting the Target Variable2
 - 2.4. Models used in our Project.....3
 - 2.4.1 Random Forest.....3
 - 2.4.2. Gradient Boosting.....3
 - 2.4.3. XGBoost3
 - 2.4.4. LightGBM.....3
 - 2.5. Evaluation Metrics used in our Project4
 - 2.6. Summary of Literature Review.....4
- 3. METHODOLOGY5
 - 3.1. Data Collection.....5
 - 3.1.1. Dataset Description.....5
 - 3.1.2. Data Characteristics5
 - 3.1.3. Exclusion Criteria6
 - 3.2. Data Preprocessing.....6
 - 3.2.1. Separation of Numerical and Categorical Attributes.....7
 - 3.2.2. Cleaning of Categorical Features.....7
 - 3.2.3. Cleaning of Numerical Features8
 - 3.3. Feature Collection8
 - 3.3.1 Method8
 - 3.3.2 Rationale9

4. RESULTS	10
4.1. Model training and testing	10
4.1.1. Training Process.....	10
4.1.2. Testing Process	10
4.2. Distribution Plots	11
4.2.1. The correlation between "Duration" and Target y	12
4.2.2. The correlation between “euribor3m’ and Target y	12
4.2.3. The correlation between “cons.price.idx” and Target y	13
4.2.4. The correlation between “cons.conf.idx” and Target y	14
4.2.5. The correlation between “nr.employed” and Target y	14
4.3. Performance Evaluation.....	15
4.3.1. Metric scores.....	15
4.3.2. Confusion Matrix and ROC Curve Analysis	16
4.3.3. Feature Importance	18
4.4. Analysis of Results.....	22
4.4.1. Interpretation of Model Performance	22
4.4.2. Significant Findings.....	23
4.5. Challenges and Limitations.....	23
5. CONCLUSION AND FUTURE WORK.....	25
5.1. Conclusion.....	25
5.2. Future work	25
5.3. Recommendations for Bank.....	26
REFERENCES	27

List of Figures

Figure 4.1: Influence of all 20 raw features on the Target y	11
Figure 4.2: Call Duration vs Subscription Outcome.....	12
Figure 4.3: Euribor3m vs Subscription Outcome	12
Figure 4.4: Cons.price.idx vs Subscription Outcome	13
Figure 4.5: Cons.conf.idx vs Subscription Outcome	14
Figure 4.6: Nr. Employed vs Subscription Outcome	14
Figure 4.7: Confusion matrix and ROC curve of the Random Forest model	16
Figure 4.8: Confusion matrix and ROC curve of the XGBoost model.....	17
Figure 4.9: Confusion matrix and ROC curve of the Gradient Boosting model	17
Figure 4.10: Confusion matrix and ROC curve of the LightGBM model	18
Figure 4.11: Feature importance of the Random Forest model	19
Figure 4.12: Feature importance of the XGBoost model.....	20
Figure 4.13: Feature importance of the Gradient Boosting model	21
Figure 4.14: Feature importance of the LightGBM model	22

List of Tables

Table 4.1. Performance comparison of classification models on the test set..... 11

Table 4.2. Overfitting analysis table 15

CHAPTER 1

INTRODUCTION

1.1. Background

In recent years, the rapid expansion of data-driven technologies has transformed how organizations make decisions, especially in fields where understanding patterns and predicting outcomes are crucial. As datasets grow and complex, traditional analytical methods often become insufficient, creating a strong need for more advanced machine learning approaches. These models not only help uncover hidden relationships within the data but also provide reliable predictions that support strategic planning and operational improvements [3].

1.2. Problem statement

In this project, we focus on building classification models to predict binary outcomes, using a set of features that have been pre-processed and encoded into numerical forms. The main objective is to evaluate how different machine learning algorithms perform on the same dataset and determine which model offers the most accurate and stable results. This involves comparing both linear and ensemble-based methods to capture a wide range of modeling behaviors.

1.3. Research objective

To achieve this, we employ five widely used classification algorithms: Random Forest, Gradient Boosting, XGBoost, and LightGBM. These models were selected because they represent diverse learning mechanisms, from straightforward linear decision boundaries to highly flexible tree-based ensemble techniques capable of capturing complex nonlinear interactions. Evaluating multiple models allows us to understand their strengths, weaknesses, and suitability for the given prediction task.

1.4. Scope of study

Model performance is assessed using commonly accepted classification metrics, including Accuracy, Precision, Recall, F1-score, and ROC-AUC [6]. Each metric offers a different performance, helping us avoid biased conclusions that could arise from relying on a single measure. By combining these evaluation criteria, we aim to provide a well-rounded and reliable comparison of the models.

1.5. Organization of the report

Overall, this study aims to deliver insights into which machine learning techniques are most effective for binary prediction problems in structured tabular data. The findings may serve as a reference for future research and real-world applications where classification accuracy and reliability are essential.

CHAPTER 2

LITERATURE REVIEW

2.1. Machine Learning for classification problems

In recent years, machine learning has become one of the most widely used approaches for building predictive models across many fields, ranging from healthcare and finance to marketing and social sciences. Among the wide spectrum of available algorithms, classification models play an especially important role when the goal is to predict binary outcomes, such as whether a customer will make a purchase, whether a loan will default, or whether a house price will exceed a certain threshold. This study focuses on several widely adopted classification models that represent both traditional statistical techniques and modern ensemble-learning methods.

2.2. Synthesis of Relevant Datasets

Numerous studies in the field of bank marketing prediction have relied on publicly available datasets that capture customer demographics, financial behaviors, and marketing interaction histories. One of the most widely used datasets is the Bank Marketing Dataset provided by the University of California Irvine (UCI) Machine Learning Repository [1][2]. This dataset includes detailed information about customer age, job category, marital status, education level, default history, and previous campaign outcomes.

Other related datasets include synthetic or institution-specific marketing records collected by financial institutions for analyzing customer retention, and loan approval. These datasets commonly share similar characteristics:

- A mixture of numerical (e.g., age, duration) and categorical features (e.g., job, marital, contact type).
- Behavioral information such as call duration, number of contacts and campaign participation.
- A binary target variable describing customer response (e.g., subscribed/not subscribed).

Overall, these datasets provide a rich foundation for benchmarking machine learning models and comparing predictive performance across studies in marketing analytics.

2.3. Challenges in Predicting the Target Variable

Predicting customer subscription behavior presents several analytical challenges:

- **Class Imbalance**

In most bank marketing datasets, the proportion of customers who subscribe ($y=1$) is significantly lower than those who do not subscribe ($y=0$). This imbalance results in biased models that favor the majority class, making accurate prediction of the minority class more difficult.

- **Categorical Feature Complexity**

Many categorical variables (job, education, contact method) have numerous levels and exhibit nonlinear relationships. This increases the dimensionality after One-hot encoding [6] and requires models capable of handling high dimensional data.

- **Behavioral Noise**

Some features, such as duration or number of contacts, are highly variable across clients. Customer behavior during marketing campaigns may introduce noise, reducing model stability and generalization.

- **Temporal and Campaign Effects**

Marketing outcomes depend heavily on economic cycles, campaign timing, and communication strategy. Without explicit temporal modeling, traditional ML algorithms may struggle to capture these underlying patterns.

- **Feature Interaction Complexity**

Customer decision-making is influenced by interactions between socioeconomic factors and previous campaign results. Linear models may fail to capture these interactions without feature engineering.

2.4. Models used in our Project

2.4.1. Random Forest

Random Forest [4] is an ensemble learning method based on bagging (Bootstrap Aggregating). It constructs a large number of decision trees using random subsets of data and features, then aggregates their predictions. This approach improves robustness, reduces overfitting, and handles nonlinear relationships effectively. Numerous studies have shown that Random Forest provides strong baseline performance across classification tasks with minimal parameter tuning.

2.4.2. Gradient Boosting

Gradient Boosting [5] is a boosting technique that builds trees sequentially, where each new tree corrects the errors of the previous ones. It optimizes a differentiable loss function using gradient descent, allowing the model to learn complex patterns. Early implementations such as Gradient Boosting Machines (GBM) demonstrated strong predictive accuracy but were computationally expensive due to their sequential nature.

2.4.3. XGBoost

XGBoost [5] is an advanced version of Gradient Boosting, optimized for speed and performance. It introduces regularization, parallel processing, tree pruning, and efficient handling of missing values. XGBoost has become one of the most influential models in classification and regression competitions, consistently outperforming many traditional approaches.

2.4.4. LightGBM

LightGBM [5] is a gradient boosting framework developed by Microsoft, designed to be faster and more efficient than traditional GBM methods. It uses techniques such as Gradient Based One Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) to reduce computational cost. LightGBM is particularly effective on large datasets with high feature dimensionality and is widely used in industrial machine learning pipelines.

2.5. Evaluation Metrics used in our Project

- **Accuracy**

Accuracy measures the proportion of correctly classified samples. It is intuitive but may be misleading when dealing with imbalanced datasets.

- **Precision**

Precision measures how many of the predicted positive samples are actually positive. It is crucial in applications where false positives are costly.

- **Recall**

Recall measures the proportion of actual positive samples that are correctly identified. It is important in domains where missing positive cases has serious consequences, such as medical diagnosis.

- **F1-Score**

F1Score is the harmonic mean of Precision and Recall. It provides a balanced metric when the class distribution is imbalanced or when both false positives and false negatives are important.

- **ROC-AUC**

The ROC-AUC score measures the model's ability to distinguish between classes across various thresholds. AUC values close to 1 indicate strong discriminative power. This metric is commonly used in binary classification tasks to evaluate model robustness.

2.6. Summary of Literature Review

Across prior studies on bank telemarketing datasets, researchers consistently highlight the challenge of predicting subscription outcomes due to class imbalance [2], high feature heterogeneity, and the complex interaction between customer demographics, campaign conditions, and macro economic variables. Works using logistic regression, decision trees, and ensemble models (e.g., Random Forest, XGBoost) generally agree that campaign-related features, particularly *duration*, *pdays*, *previous*, and *poutcome* [2], are the strongest predictors of success. Customer attributes such as *age*, *job*, and *education* play a secondary but meaningful role, whereas economic indicators like *euribor3m* and *emp.var.rate* provide additional and *emp.var.rate* provide additional predictive value, especially during financial fluctuations. Compared to literature, our dataset exhibits similar patterns: campaign features remain the most influential, demographic attributes contribute moderate predictive power, and macro-economic variables help improve model stability. Overall, prior research and our dataset collectively emphasize that effective telemarketing prediction requires models capable of handling nonlinear relationships and imbalanced outcomes while leveraging the strongest campaign related signals.

CHAPTER 3

Methodology

This study follows a structured machine learning workflow to analyze the Bank Marketing dataset and predict customer subscription behavior. Data preprocessing included handling missing values, encoding categorical features, and scaling numerical variables. Exploratory Data Analysis (EDA) was conducted to understand feature distributions and relationships with the target variable y .

Multiple models, including Random Forest, LightGBM, Gradient Boosting and XGBoost, were trained and evaluated using stratified train, test splitting to address class imbalance. Model performance was assessed using accuracy, precision, recall, F1-score, and ROC-AUC. Hyperparameter tuning was applied to optimize final performance.

This systematic approach ensures reliable insights into the factors influencing marketing campaign success and supports robust predictive modeling.

3.1. Evaluation Metrics used in our Project

3.1.1. Dataset Description

The dataset used in this study is the Bank Marketing Dataset, originally provided by the UCI Machine Learning Repository. It contains 41,188 rows and 21 input features, each representing customer characteristics or details about a direct marketing campaign conducted by a Portuguese bank. The primary objective of the dataset is to predict whether a client will subscribe to a term deposit, represented by the binary variable y .

3.1.2. Data Characteristics

The dataset consists of both numerical and categorical features that represent customer demographics, financial behavior, marketing interaction history, and macroeconomic conditions. These attributes collectively describe each client and support the prediction of whether they will subscribe to a term deposit (target variable y).

Typical Numerical Features

- *Age*: Customer's age, representing demographic differences.
- *Balance*: Average yearly account balance, reflecting financial stability.
- *Duration*: Length of the last contact (seconds); highly influential for campaign success.
- *Campaign*: Number of contacts performed during the current campaign.
- *Pdays*: Days passed since the client was last contacted in a previous campaign.
- *Previous*: Number of contacts performed before the current campaign.
- *Emp.var.rate*: Quarterly employment variation rate.
- *Cons.price.idx*: Consumer price index.
- *Cons.conf.idx*: Consumer confidence index.
- *Euribor3m*: 3month EURIBOR rate.

- *Nr.employed*: Number of employees in the economy.

The Typical Categorical Features

- *Job*: Type of occupation (e.g., admin, technician, services).
- *Marital*: Marital status of the customer.
- *Education*: Highest educational degree attained.
- *Default*: Whether the customer has credit in default.
- *Housing*: Whether the customer has a housing loan.
- *Loan*: Whether the customer has a personal loan.
- *Contact*: Contact communication type (cellular or telephone).
- *Month*: Month of the last contact.
- *Day_of_week*: Day of the week of the last contact.
- *Poutcome*: Outcome of previous marketing campaigns.
- *Y (Target)*: Binary indicator of term deposit subscription (yes/no).

The dataset includes missing or inconsistent values that require proper preprocessing. Numerical fields are standardized or normalized as needed, while categorical features are encoded using onehot or ordinal encoding depending on model suitability. This combination of demographic, behavioral, and macroeconomic attributes provides a comprehensive foundation for predicting customer subscription decisions.

3.1.3. Exclusion Criteria

To ensure data quality and maintain the integrity of the analysis, several exclusion criteria were applied during dataset preprocessing. First, records containing critical missing information, such as the target variable *y* or essential campaign details (e.g., *duration*, *contact*, *pdays*), were removed, as their absence would compromise model learning. Second, entries with invalid or logically inconsistent values (e.g., negative durations, impossible economic indicator values) were excluded to avoid introducing noise into the predictive modeling pipeline.

Additionally, customers who were never previously contacted but had non-zero *pdays* values were filtered out due to inconsistencies with the dataset's definition. Rare categorical levels with extremely low frequency were also excluded or merged to prevent instability during model training.

These exclusion steps ensured that the final dataset was clean, consistent, and reliable, forming a solid foundation for downstream feature engineering and machine learning tasks.

3.2. Data Preprocessing

Data preprocessing was conducted to prepare the dataset effectively for machine learning models. This step involved addressing missing values, encoding categorical features, scaling numerical features, selecting relevant predictors, and managing class imbalances. The following outlines the preprocessing pipeline in detail.

- **Data cleaning**

A comprehensive data cleaning procedure was conducted to ensure that the dataset was analytically robust and ready for subsequent modeling tasks. Because the dataset contains a diverse mix of numerical, categorical, campaign related, and macroeconomic variables, a structured cleaning pipeline was developed to systematically address data quality issues such as missing values, inconsistent categories, and non-informative attributes.

3.2.1. Separation of Numerical and Categorical Attributes

To facilitate appropriate preprocessing, all variables were first divided into numerical and categorical groups:

- **Numerical features**

Such as age, duration, campaign, pdays, euribor3m, and nr.employed, were evaluated for outliers, distribution irregularities, and missing observations. Their continuous nature required different imputation and validation methods compared to categorical fields.

- **Categorical features**

Including job, marital, education, contact, month, day_of_week, and poutcome, were assessed for category inconsistencies, rare levels, and placeholder values (e.g., “unknown”). These variables capture customer characteristics and interaction patterns that are highly relevant for subscription behavior.

This separation allowed the cleaning procedures to be more precise and aligned with the statistical properties of each feature type.

3.2.2. Cleaning of Categorical Features

Because categorical variables are widely used in bank marketing models, their quality directly affects the reliability of predictive outcomes. Therefore, an indepth cleaning workflow was applied to these attributes.

- **Handling Missing Values**

Missing values in categorical attributes were common, often represented as "unknown". To avoid losing samples and preserve distributional integrity:

Mode imputation was applied to fill missing entries.

For features with a high prevalence of “unknown”, the category was retained as a meaningful representation of uncertainty, since prior studies indicate that “unknown” sometimes carries behavioral information in bank marketing campaigns.

This ensured that categorical variables remained complete while maintaining interpretive consistency.

- **Dropping Irrelevant or Low-Quality Columns**

Certain features were identified as non-informative or detrimental to modeling quality:

Variables with extremely sparse categories (appearing in less than ~1% of observations) were removed to avoid noise amplification.

Columns with inconsistent definitions across records or redundant information were eliminated to reduce multicollinearity.

Campaign metadata that did not contribute to predictive performance (e.g., identifiers, technical fields) were excluded.

This step streamlined the feature space, improved model convergence, and reduced the risk of overfitting.

- **Encoding Categorical Features Using Column Transformer and Pipelines**

Instead of applying simple label encoding directly on the dataset, the categorical attributes were encoded within a unified preprocessing pipeline validation using the Column Transformer [6] module from scikitlearn. In this setup, numerical and categorical features were processed in parallel: numerical variables were handled by a subpipeline including median imputation and Min–Max normalization, while categorical variables were passed through a dedicated subpipeline combining mode imputation with categorical encoding (e.g., one hot or ordinal encoding, depending on the model requirements).

By embedding the encoding step inside a Column Transformer and attaching it to a full modeling pipeline, all transformations were fitted only on the training data and then consistently applied to the validation and test sets, thereby preventing data leakage. This design also ensured reproducibility, cleaner code structure, and seamless integration with cross-validation [6] and hyperparameter tuning procedures used in the subsequent model training phase.

3.2.3. Cleaning of Numerical Features

For numerical variables, the preprocessing pipeline focused on ensuring consistency, reducing scale-related bias, and preserving the interpretability of each attribute. Numerical features were first examined for missing values; however, since the dataset contained no missing entries in this group, no imputation methods were required. Following this verification, the data underwent Min–Max normalization to rescale all numerical attributes into the range [0,1]

This transformation prevents features with larger numeric ranges, such as *duration*, *campaign*, or economic indicators, from disproportionately influencing the model during training. By compressing all values into a uniform scale, Min–Max normalization enhances model stability and ensures fair contribution from each numerical feature. This standardized representation ultimately supports improved training convergence and more balanced predictive performance across classification models.

3.3. Feature Collection

3.3.1. Method

The feature collection process involved systematically extracting all relevant variables from the original Bank Marketing dataset provided by the UCI Machine Learning Repository [1]. Both customer-related attributes (e.g., *age*, *job*, *marital status*), marketing campaign variables (e.g., contact method, previous campaign outcome), and macroeconomic indicators (e.g., employment variation rate, consumer confidence index) were included. Features were grouped into three main categories: customer demographics, campaign-related characteristics, and macroeconomic indicators, to support structured downstream analysis.

To ensure data integrity during collection, only variables directly present in the original dataset were retained. No synthetic, engineering, or external features were added during this stage. Each attribute was

carefully examined to verify data type consistency, completeness, and relevance before proceeding to subsequent preprocessing steps.

3.3.2. Rationale

The rationale behind this feature collection strategy is to preserve the full informational structure of the data while enabling a comprehensive analysis of the factors influencing the subscription outcome (y).

- **Customer features** were collected because demographic and financial characteristics often strongly correlate with consumer decision making behavior.
- **Marketing campaign features** were retained because interaction history, contact frequency, and campaign success metrics have direct predictive value for customer response.
- **Macroeconomic indicators** were included as they capture broader economic conditions that may influence customers' willingness or ability to subscribe to a long-term deposit product.

By collecting all raw features prior to transformation or reduction, the analysis avoids premature information loss and ensures that later feature selection techniques operate on a complete and unbiased feature space. This approach aligns with best practices in supervised machine learning workflows, where initial feature completeness is essential for accurate model development.

CHAPTER 4

RESULTS

4.1. Model training and testing

4.1.1. Training process

The model training process was designed to ensure fair evaluation, avoid data leakage, and achieve reliable predictive performance. After preprocessing, the dataset was split into 80% training and 20% testing using stratified sampling to preserve the original class distribution. To enhance generalization and reduce overfitting, all models were trained with 5-fold cross-validation, and performance was assessed using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

For Random Forest, Gradient Boosting, XGBoost, and LightGBM, hyperparameter tuning was performed within the cross-validation framework to identify optimal configurations. All preprocessing steps, including categorical encoding and numerical scaling, were implemented in a unified pipeline to prevent leakage and ensure methodological consistency. Class-weight balancing was also applied to address class imbalance and improve minority-class detection.

Overall, this structured pipeline ensured consistent, controlled, and reproducible model evaluation, providing a strong foundation for the performance comparisons presented in the Results section.

4.1.2. Testing process

After model training and hyperparameter optimization, all finalized models were evaluated on a held-out test set that was reserved during the initial stratified train–test split. This test set was excluded from all stages of model fitting and tuning, ensuring that the reported results provide an unbiased estimate of each model’s generalization performance on unseen data and reduce the risk of overfitting and information leakage.

Model performance was assessed using multiple metrics, including Accuracy, Precision, Recall, F1-score, and ROC-AUC, to capture different aspects of predictive performance under class imbalance. While Accuracy reflects overall correctness, Precision and Recall describe the trade-off between false positive and false negative predictions. The F1-score was emphasized as a key metric because it balances Precision and Recall, which is particularly important for the term deposit subscription task where both types of errors have practical consequences. ROC-AUC was additionally used to evaluate the models’ discriminative ability across different decision thresholds.

Table 4.1. Performance comparison of classification models on the test set

	Model	Accuracy	Precision	Recall	F1	ROC_AUC
1	Random Forest	0.8874	0.5000	0.5000	0.5241	0.8069
2	Gradient Boosting	0.8769	0.4629	0.5776	0.5139	0.8000
3	XGBoost	0.8768	0.4634	0.5938	0.5206	0.8034
4	LightGBM	0.8879	0.5024	0.5614	0.5303	0.8058

Overall, according to Table 4.1, the boosting-based models, particularly LightGBM, demonstrated stronger performance in terms of F1-score and Recall compared to Random Forest. This indicates their superior ability to capture nonlinear relationships and handle class imbalance in the term deposit subscription prediction task.

4.2. Distribution Plots

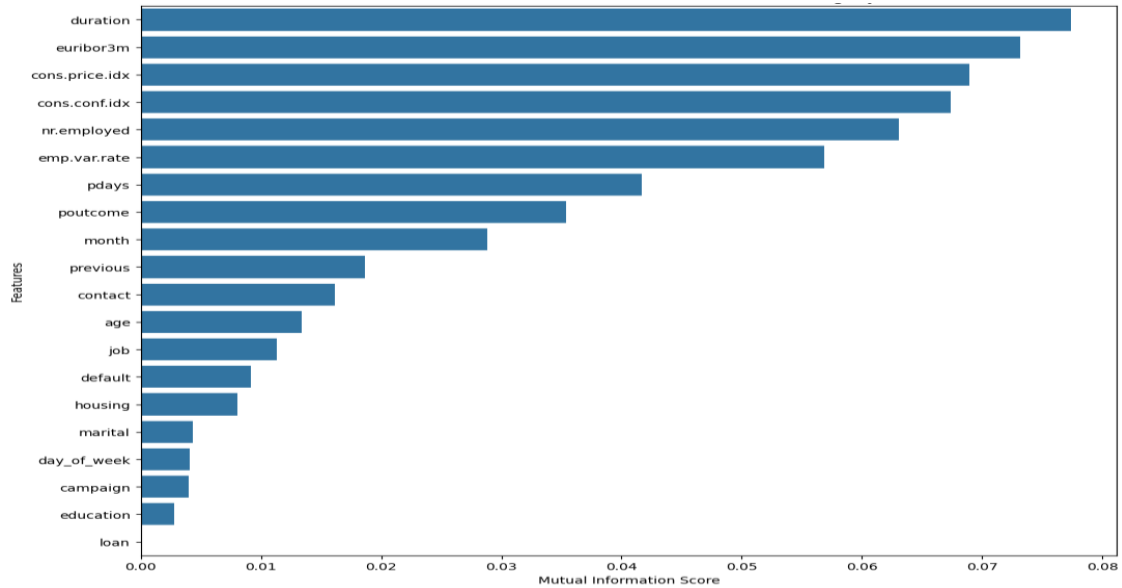


Figure 4.1: Influence of all 20 raw features on the Target.

Fig. 4.1 presents the Mutual Information (MI) [7] results, showing how strongly each feature relates to the target subscription outcome. The feature *duration* has the highest MI, indicating a very strong relationship; however, since it is only known after the call, it causes data leakage and should not be used in real predictive models. Several macroeconomic variables (*euribor3m*, *cons.price.idx*, *cons.conf.idx*, *nr.employed*, *emp.var.rate*) also rank highly, suggesting that economic conditions significantly influence customers' decisions. Campaign-related features (*pdays*, *poutcome*, *previous*, *month*) show moderate importance, reflecting the impact of past interactions, while demographic variables such as age, job, and education contribute relatively less. Overall, MI highlights that behavioral and economic features are

more informative than demographics, and leakage-prone features like duration must be excluded to ensure realistic model performance

4.2.1. The correlation between “Duration” and Target

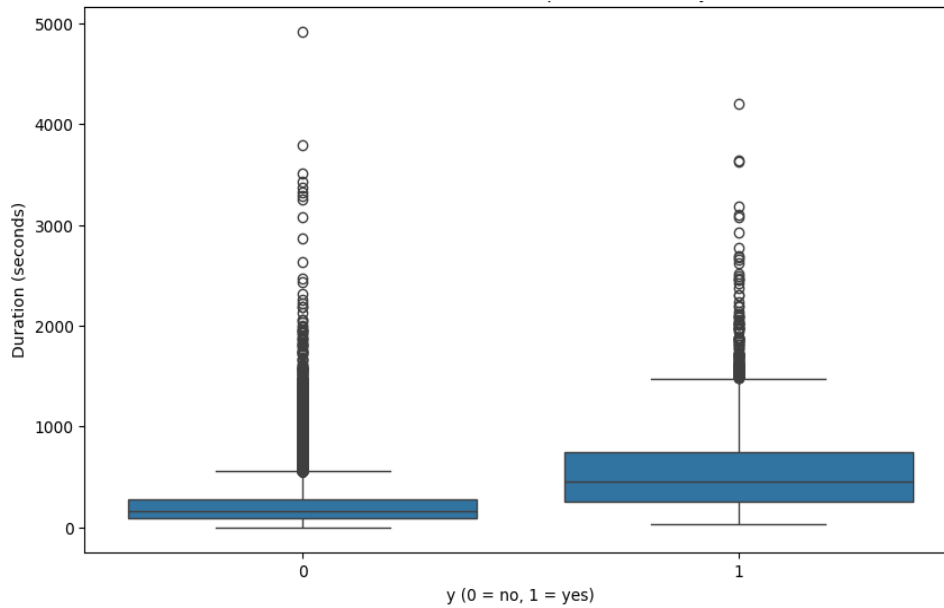


Figure 4.2: Call Duration vs Subscription Outcome.

Fig. 4.2 shows a clear difference between the two groups. Customers who subscribed (“yes”) generally had much longer call durations than those who did not (“no”). The “yes” group has a noticeably higher median duration (around ~500-600 seconds) compared to the “no” group (around ~150-200 seconds). The “yes” group also shows a wider spread and many longer calls, with several calls exceeding 3000-4000 seconds, while most “no” calls remain relatively short. Overall, this indicates that longer conversations are strongly associated with a higher likelihood of subscription.

4.2.2. The correlation between “euribor3m” and Target

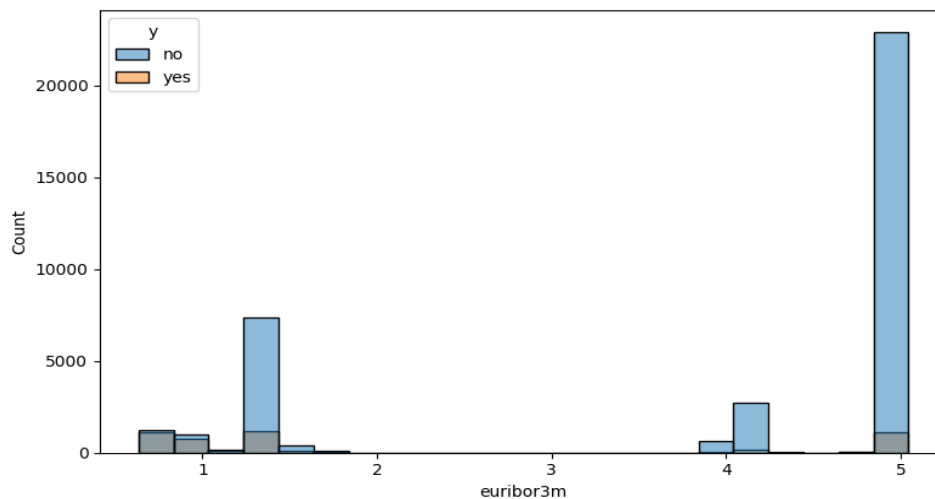


Figure 4.3: Euribor3m vs Subscription Outcome.

Fig. 4.3 shows a clear difference in *euribor3m* values between customers who subscribed (“yes”) and those who did not (“no”). Customers in the “yes” group generally have much lower *euribor3m* rates, with the median around ~1.2, compared to the “no” group, where the median is close to ~4.9-5.0. The “no” group is heavily concentrated at higher interest rate levels, while the “yes” group is more spread but centered around lower values. Overall, this indicates that customers are more likely to subscribe when *euribor3m* rates are low, suggesting that favorable economic conditions (lower interest rates) encourage term deposit subscriptions.

4.2.3. The correlation between “cons.price.idx” and Target

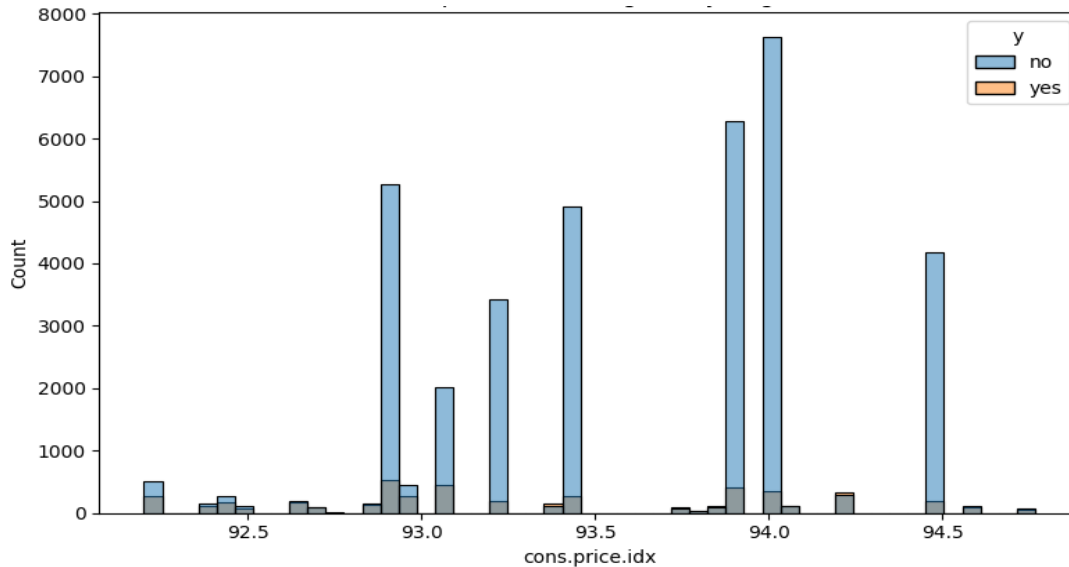


Figure 4.4: *Cons.price.idx* vs Subscription Outcome.

Fig. 4.4 compares consumer price index (*cons.price.idx*) between customers who subscribed (“yes”) and those who did not (“no”). The difference is subtle but noticeable. The “yes” group tends to have a slightly lower median *cons.price.idx* (around ~93.2-93.4) compared to the “no” group, where the median is closer to ~93.8-94.0. Both distributions overlap significantly, but the shift suggests that customers are slightly more likely to subscribe when the consumer price index is lower, implying that more stable or less inflationary economic periods may encourage term deposit subscriptions.

4.2.4. The correlation between “cons.conf.idx” and Target

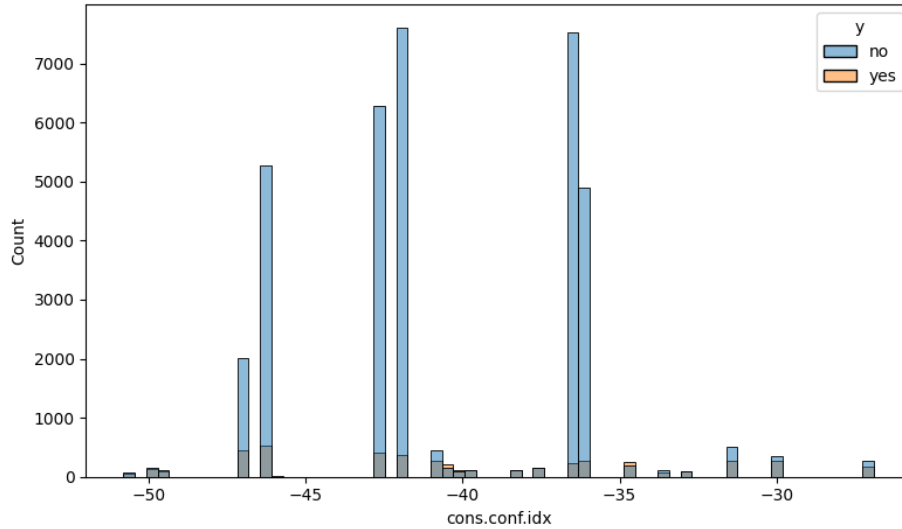


Figure 4.5: *Cons.conf.idx* vs Subscription Outcome.

Fig. 4.5 compares the consumer confidence index (*cons.conf.idx*) between customers who subscribed (“yes”) and those who did not (“no”). Overall, the two groups show similar distributions, but there is a slight shift downward for the “yes” group. The median consumer confidence for non-subscribers is around -40 to -41, while for subscribers it is slightly lower (around -42 to -44). The “yes” group also shows a slightly wider spread toward lower confidence values. This suggests that customers may be more likely to subscribe during periods of lower consumer confidence, possibly because uncertainty encourages people to secure their savings in safer financial products like term deposits.

4.2.5. The correlation between “nr.employed” and Target y

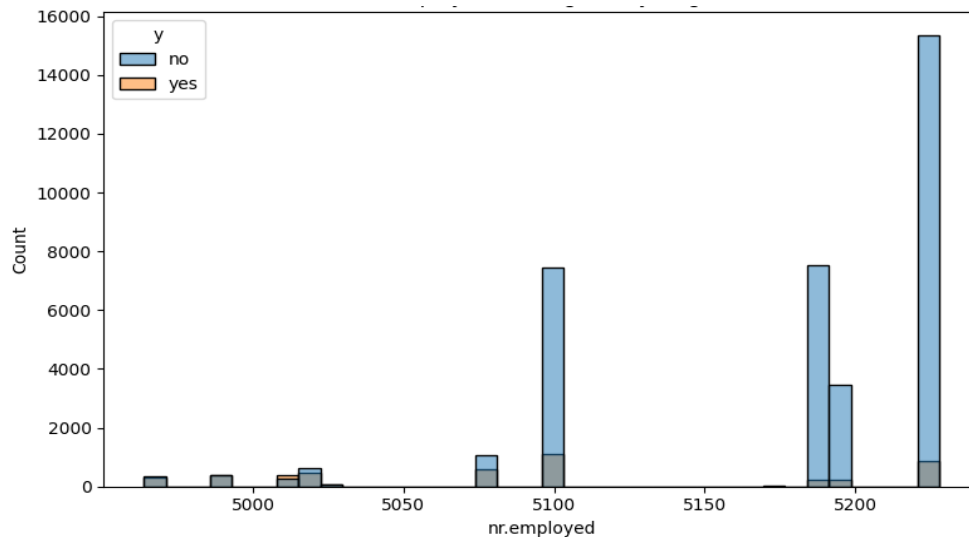


Figure 4.6: *Nr. Employed* vs Subscription Outcome.

Fig. 4.6 compares *nr.employed* (number of employees in the economy) between customers who subscribed (“yes”) and those who did not (“no”). Overall, the non-subscriber group (“no”) has a higher median value, around ~5200, while the subscriber group (“yes”) has a lower median, roughly ~5100. The

distribution for “yes” customers also spans slightly lower employment levels. Fig. 4.6 suggests that term deposit subscriptions are more likely when employment levels are lower, possibly reflecting periods of weaker economic conditions. In such contexts, customers may prefer safer, guaranteed-return products like term deposits, leading to higher subscription rates.

4.3. Performance Evaluation

The performance metrics reported in the notebook include Accuracy, Precision, Recall, F1-score, and ROC-AUC, which were used to evaluate the ensemble-based models presented in the results table. The comparison focuses exclusively on Random Forest, Gradient Boosting, XGBoost, and LightGBM, as these are the models for which test-set performance metrics were consistently computed and aggregated. These metrics provide a comprehensive view of model performance. Accuracy measures overall classification correctness, while Precision and Recall evaluate the model’s effectiveness in identifying subscribing customers. The F1-score, by jointly considering Precision and Recall, offers a balanced assessment of predictive performance, particularly under class imbalance. In addition, ROC-AUC reflects the models’ discriminative capability across different decision thresholds.

4.3.1. Metric Scores

The notebook generated a comprehensive set of performance metrics to evaluate each trained model. As summarized in Table 4.1, five evaluation metrics were reported, including Accuracy, Precision, Recall, F1-score, and ROC-AUC. These metrics were selected to account for the class imbalance present in the dataset and to provide a balanced assessment of both overall classification correctness and the models’ ability to identify subscribing customers.

The results table presents the test-set performance of the four ensemble-based classifiers implemented in the notebook: Random Forest, Gradient Boosting, XGBoost, and LightGBM. Overall, all models achieved relatively high Accuracy and comparable ROC-AUC values, indicating similar discriminative capability at a global level. However, clearer performance differences emerge when considering Recall and F1-score, which are more sensitive to minority-class detection.

In addition to test-set evaluation, the notebook includes an overfitting analysis comparing training and testing performance in terms of Accuracy and F1-score. The results of this analysis are reported in Table 4.2. which provides further insight into each model’s generalization behavior

Table 4.2. Overfitting analysis table

	Model	Train_Acc	Test_Acc	Train_F1	Test_F1
1	Random Forest	0.8677	0.8874	0.4965	0.5241
2	Gradient Boosting	0.8599	0.8769	0.4804	0.5139
3	XGBoost	0.8831	0.8768	0.4843	0.5206
4	LightGBM	0.90061	0.8880	0.4852	0.5303

Table 4.2 shows that although all models achieve strong training performance, differences between training and testing metrics are observed, indicating varying degrees of overfitting. LightGBM exhibit relatively smaller train-test Accuracy gaps and stable F1-score differences, suggesting stronger robustness and more reliable generalization to unseen data.

Overall, the metric scores emphasize the strong predictive capacity of the tree-based ensemble classifier. At the same time, it also highlights the significance of monitoring generalization behavior, especially for high-capacity models.

4.3.2. Confusion Matrix and ROC Curve Analysis

This subsection jointly analyzes the confusion matrices and ROC curves for each model evaluated in the notebook. The analysis is organized by model to improve clarity and traceability between quantitative metrics, error patterns, and discriminative performance.

- **Random Forest**

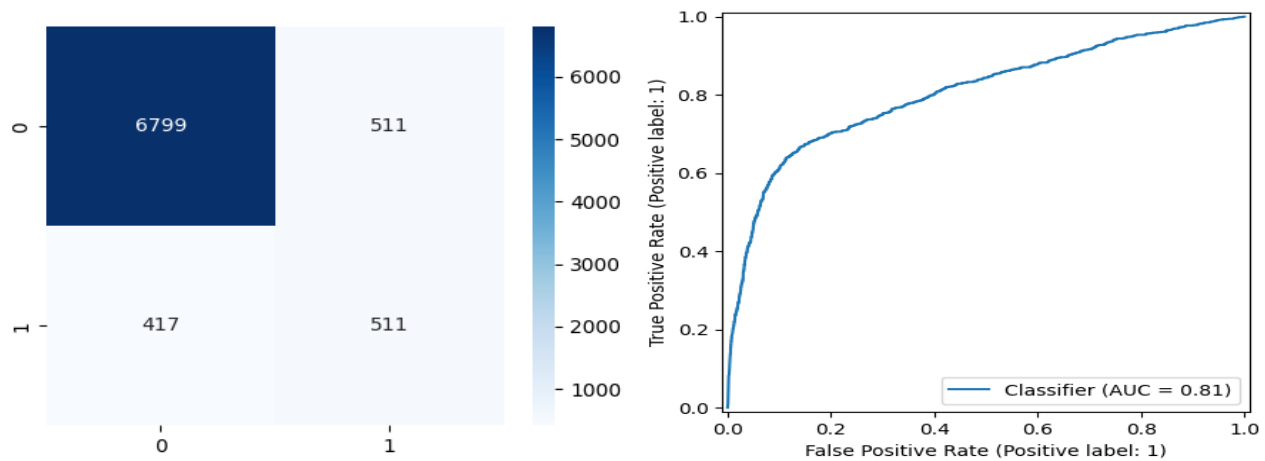


Figure 4.7. Confusion matrix and ROC curve of the Random Forest model.

Fig. 4.7 shows that the Random Forest model improves the identification of subscribing customers compared to simpler baseline models, mainly through a reduction in false negative predictions and a corresponding increase in Recall for the positive class. This indicates better detection of likely subscribers, although a notable number of false negatives remains, reflecting limitations in capturing more complex patterns. The associated ROC curve exhibits moderate discriminative ability, lying above the diagonal reference line but showing weaker class separation than boosting-based models. This is consistent with the lower ROC-AUC value reported for the Random Forest [4] model, confirming its comparatively weaker overall classification performance.

- **XGBoost**

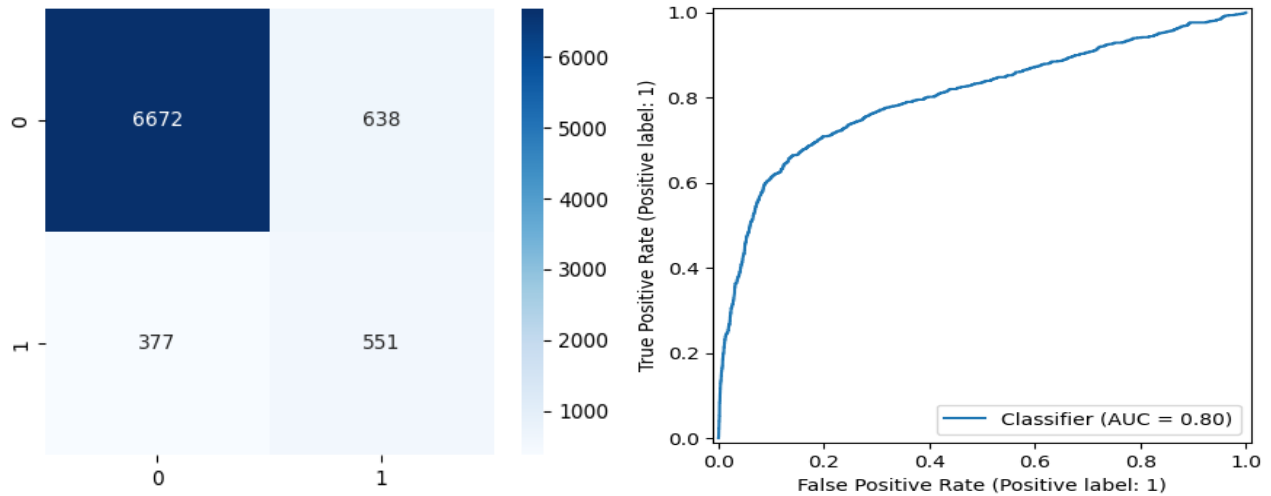


Figure 4.8. Confusion matrix and ROC curve of the XGBoost model.

Fig. 4.8 illustrates that XGBoost achieves a more balanced classification performance than the Random Forest model, with an increase in true positive predictions and a reduction in false negatives, leading to improved Recall for the positive class. While false positives increase slightly, they remain well controlled, resulting in a better trade-off between Precision and Recall and a higher F1-score. The ROC curve further confirms this improvement, as it lies consistently above that of Random Forest, indicating stronger class separation. Although the gain in ROC-AUC is moderate, it demonstrates greater robustness across classification thresholds and confirms the effectiveness of XGBoost [5] in capturing complex patterns in the data.

- **Gradient Boosting**

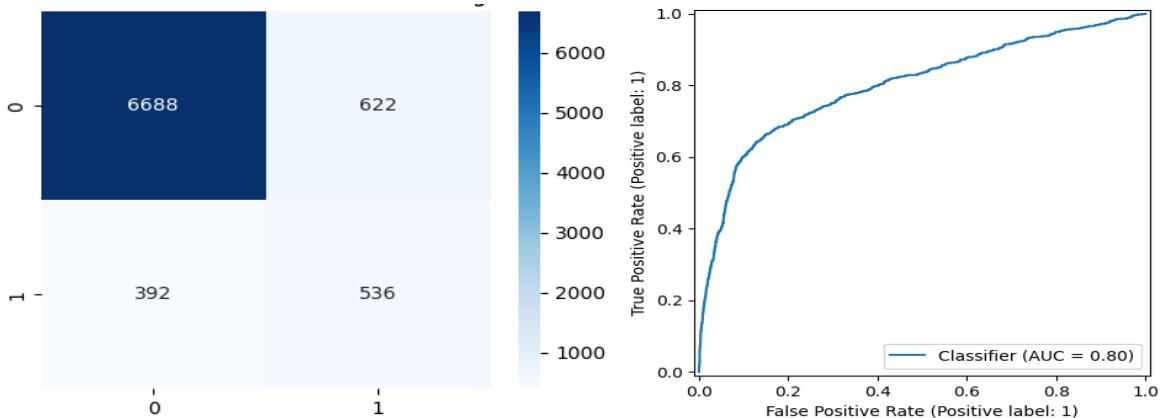


Figure 4.9. Confusion matrix and ROC curve of the Gradient Boosting model.

Fig. 4.9 shows that the Gradient Boosting model achieves a balanced distribution of prediction errors, with a relatively high number of true positives and fewer false negatives compared to simpler models, indicating improved Recall for the positive class. Although false positives remain, their proportion is moderate, reflecting a reasonable balance between sensitivity and specificity. This behavior is consistent

with the ROC curve, which lies well above the diagonal reference line and demonstrates solid discriminative ability across different thresholds. Overall, the results confirm that Gradient Boosting effectively captures non-linear relationships in the data, although its performance is comparable rather than clearly superior to the other boosting-based models considered in this study.

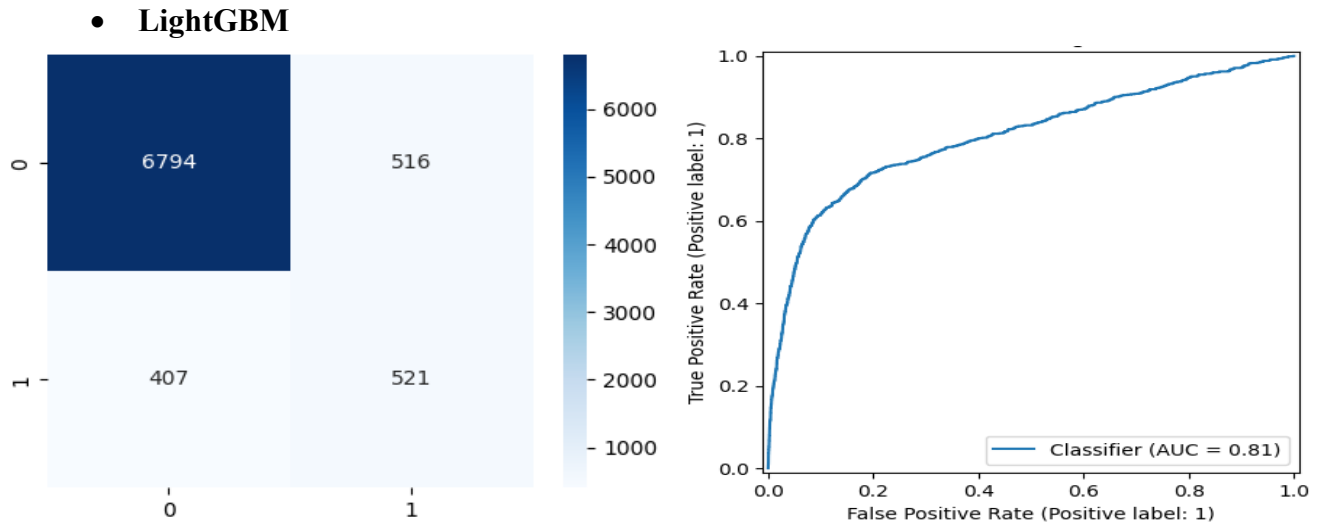


Figure 4.10. Confusion matrix and ROC curve of the LightGBM model.

Fig. 4.10 demonstrates that LightGBM achieves strong and well-balanced classification performance, correctly identifying a high number of subscribing customers while maintaining accurate predictions for non-subscribers. The low number of false negatives indicates good Recall for the positive class, while the controlled level of false positives reflects effective class discrimination, contributing to a strong F1-score. The ROC curve further confirms the model's robustness, as it consistently lies above those of several other models and maintains a high true positive rate across a wide range of false positive rates. Consequently, the high ROC-AUC value highlights LightGBM's excellent discriminative ability and stable performance across different classification thresholds.

4.3.3. Feature Importance

This subsection analyzes feature importance results for each tree-based model included in the notebook. Feature importance scores are obtained directly from the trained models and reflect the relative contribution of each feature to prediction outcomes. Only models and features explicitly used and visualized in the notebook are reported.

- **Random Forest**

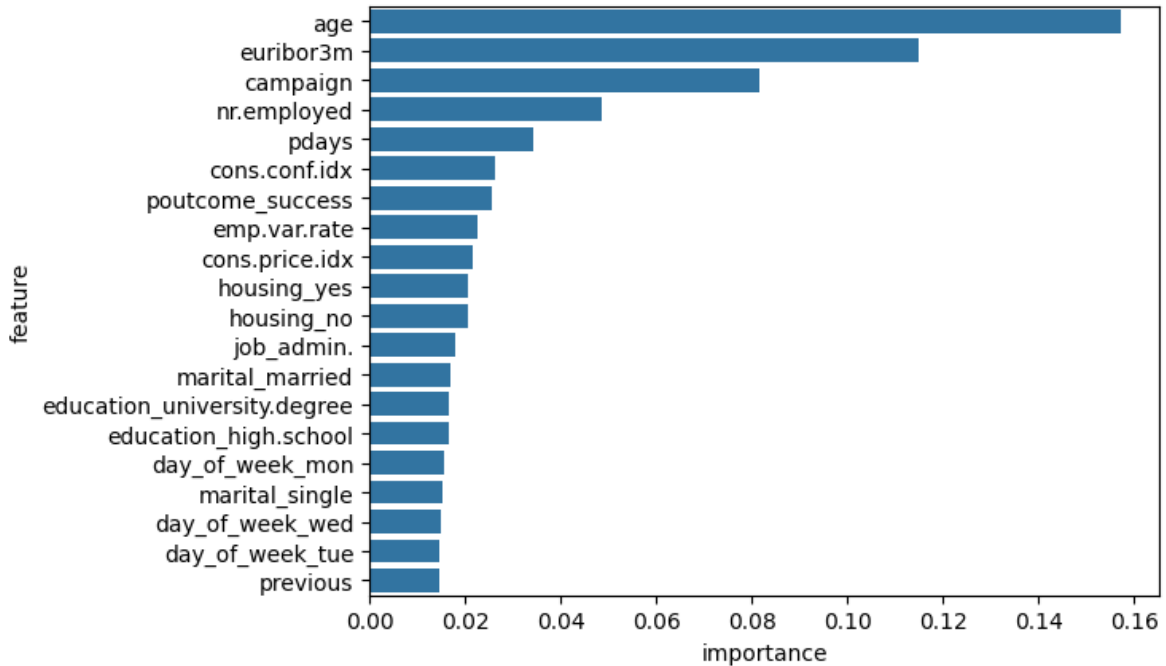


Figure 4.11. Feature importance of the Random Forest model.

Based on Fig. 4.11, age emerges as the most influential feature, indicating that customer demographic characteristics strongly affect the likelihood of subscribing to a term deposit. This suggests that subscription behavior varies across different age groups, and the model relies heavily on this attribute when making predictions.

In addition to demographic factors, several macroeconomic indicators, including *euribor3m*, *nr.employed*, and *emp.var.rate*, show high levels of importance. The prominence of these variables highlights the impact of broader economic conditions, such as interest rates and employment trends, on customer decisions. Furthermore, campaign-related features such as *campaign*, *pdays*, and *poutcome_success* contribute meaningfully, reflecting the influence of marketing intensity, recency of contact, and outcomes of previous campaigns on subscription behavior.

Other variables, including loan status, job category, education level, and day-of-week indicators, exhibit lower but non-negligible importance. Together, these features demonstrate the Random Forest model's ability to integrate demographic, behavioral, and macroeconomic information through nonlinear interactions across multiple decision trees.

- **XGBoost**

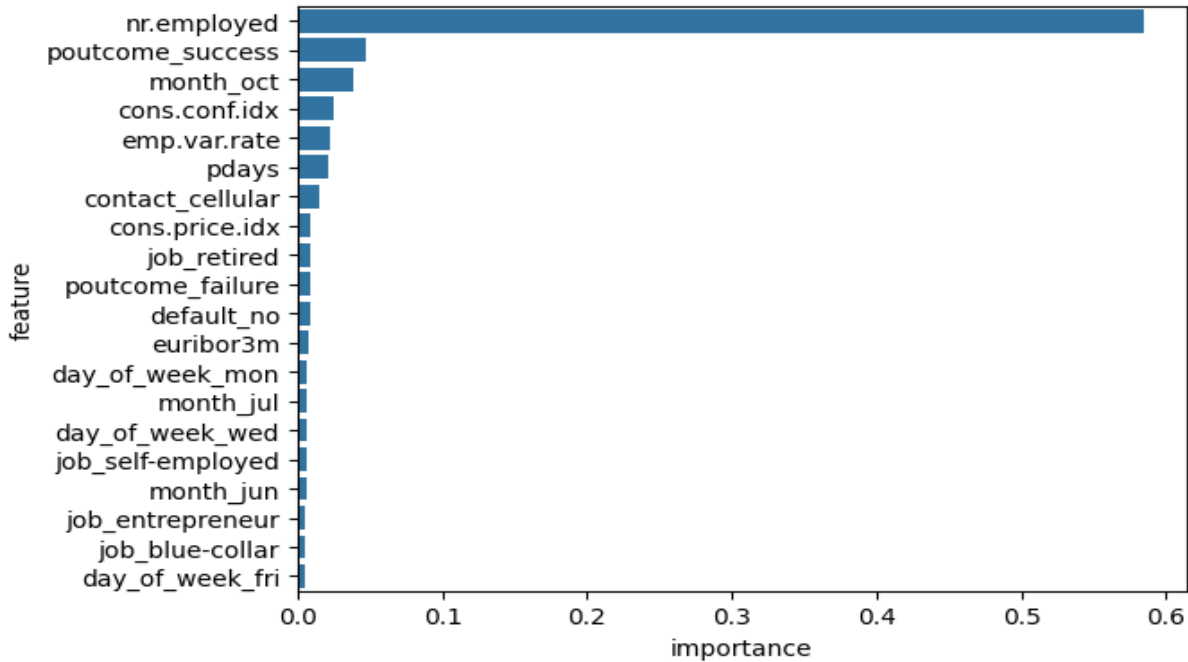


Figure 4.12: Feature importance of the XGBoost model.

Based on Fig. 4.12, *nr.employed* clearly emerges as the most influential feature by a substantial margin. This indicates that overall employment levels play a dominant role in the model's prediction of term deposit subscription. The importance distribution is highly concentrated, suggesting that XGBoost relies heavily on a small number of key features rather than distributing importance evenly across all variables.

In addition to *nr.employed*, features related to previous campaign outcomes, such as *poutcome_success*, contribute notably to the model's decisions. Temporal variables, including *month_oct*, also rank among the top features, highlighting the relevance of timing effects in customer responses. Furthermore, several macroeconomic indicators, such as *cons.conf.idx* and *emp.var.rate*, show meaningful importance, reinforcing the influence of broader economic conditions on subscription behavior. Other features related to campaign history and contact characteristics, including *pdays* and *contact_cellular*, provide additional predictive value. Overall, the feature importance pattern illustrates XGBoost's tendency to focus on a compact set of highly informative variables, leading to a more targeted and discriminative prediction strategy.

- **Gradient Boosting**

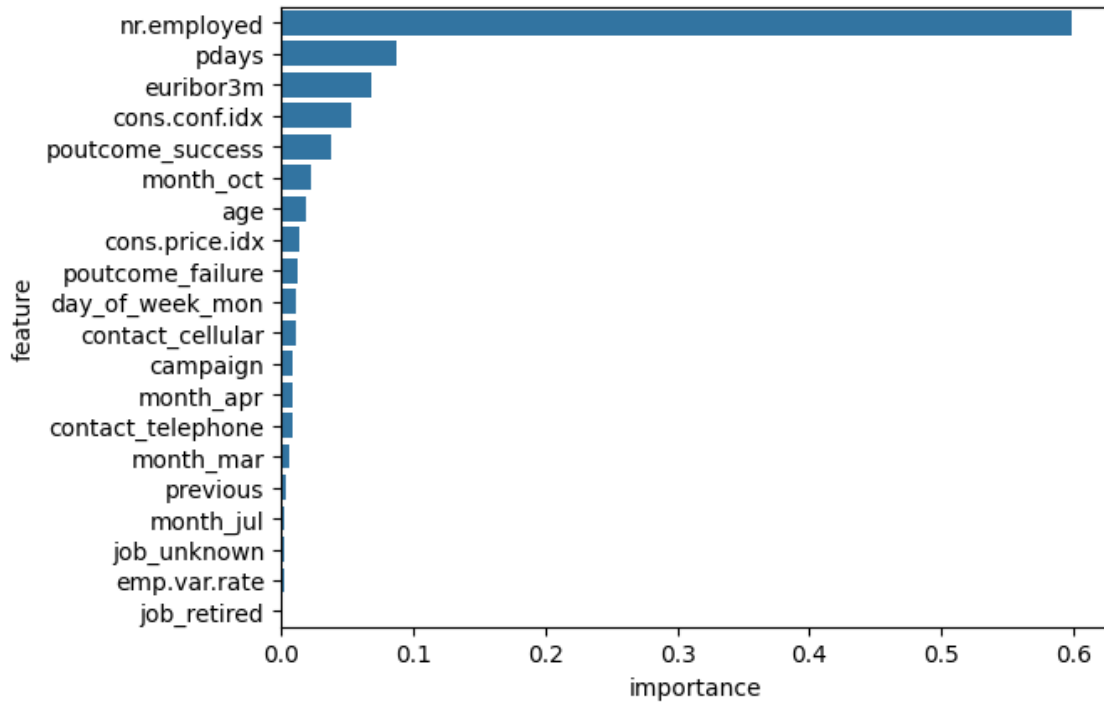


Figure 4.13. Feature importance of the Gradient Boosting model.

Based on Fig. 4.13, it is clear that *nr.employed* is by far the most influential feature, contributing a significantly larger share of the total importance compared to all other variables. This suggests that macroeconomic conditions play a crucial role in determining whether a customer is likely to subscribe. In comparison with the Random Forest model, the feature importance distribution of Gradient Boosting is notably more concentrated, implying that the predictive performance of the model is primarily driven by a limited set of key variables rather than a broad combination of features. Following *nr.employed*, variables such as *pdays*, *euribor3m*, and *cons.conf.idx* emerge as important contributors, highlighting the combined relevance of customer contact history and economic indicators.

In addition, features related to previous marketing outcomes and campaign effectiveness, such as *poutcome_success*, *poutcome_failure*, and *campaign*, remain among the top contributors. This highlights the importance of past customer interactions and responses to earlier campaigns in predicting subscription behavior. Overall, the results suggest that while Gradient Boosting leverages a wide range of features, most of its predictive power is driven by a limited number of key variables, leading to a more focused and efficient use of information compared to Random Forest.

- **LightGBM**

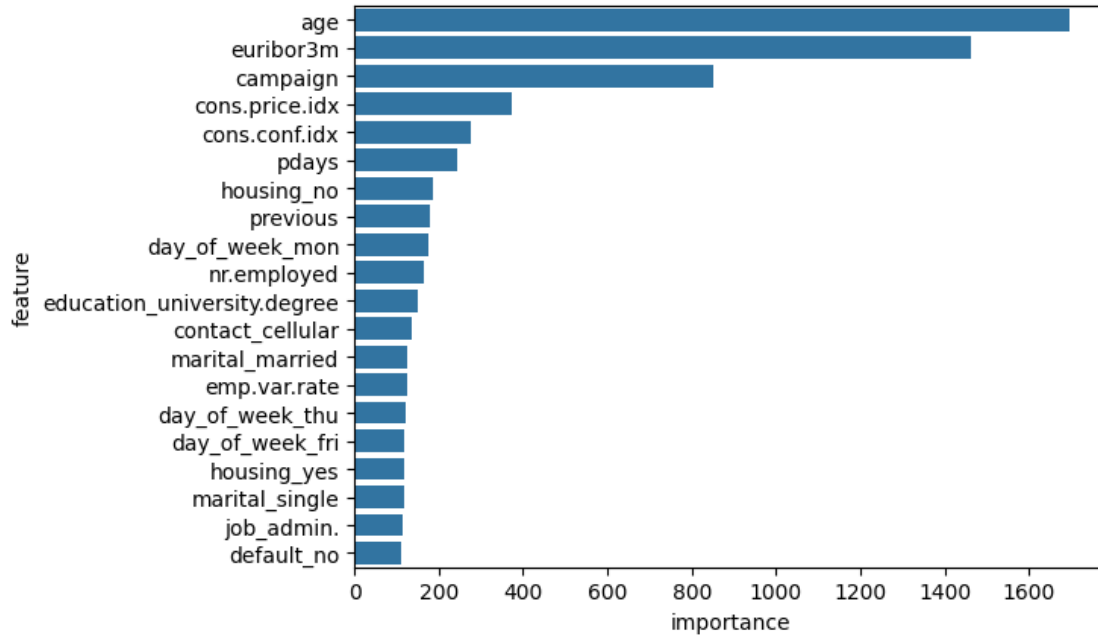


Figure 4.14. Feature importance of the LightGBM model

Fig. 4.14 presents the feature importance results obtained from the LightGBM model. The overall pattern shows strong similarities with other gradient boosting-based approaches, while also revealing some distinct characteristics. Notably, *age* emerges as the most influential feature, followed by *euribor3m* and *campaign*, indicating that both demographic attributes and economic conditions play a central role in the model's predictions.

Several macroeconomic variables, including *euribor3m*, *cons.price.idx*, *cons.conf.idx*, and *nr.employed*, rank highly, underscoring the importance of broader economic signals in explaining customer subscription behavior. At the same time, features related to marketing interactions and customer history, such as *campaign*, *pdays*, and *previous*, also contribute substantially to the predictive power of the model. This highlights LightGBM's ability to capture both short-term interaction effects and long-term contextual information.

Compared to XGBoost, the feature importance distribution in LightGBM appears to be more evenly spread across variables, suggesting a more stable learning process that does not overly rely on a single dominant feature. Overall, the results indicate that LightGBM effectively integrates demographic, interaction-based, and macroeconomic features, which helps explain its strong predictive performance as reflected in the evaluation metrics reported earlier.

4.4. Analysis of Results

4.4.1. Interpretation of Model Performance

- Clear performance differences are observed among the ensemble models.

- Boosting-based methods (XGBoost and LightGBM) achieve the highest test Accuracy and F1-scores, demonstrating strong capability in modeling non-linear relationships and decision boundaries.
- These models show smaller gaps between training and testing performance, indicating better generalization.
- Random Forest and Gradient Boosting provide reasonable baseline performance but struggle to balance Precision and Recall, resulting in higher false negative rates.
- Overall, boosting-based models deliver more robust and reliable performance, particularly under class imbalance.

4.4.2. Significant Findings

- Differences in model performance are primarily explained by how effectively each model detects the minority class (term deposit subscribers).
- Models with higher F1-scores achieve better balance between Precision and Recall, mainly by reducing false negative predictions.
- This confirms that accuracy alone is insufficient for evaluating performance in imbalanced classification problems.
- Feature importance analysis shows that marketing interaction variables dominate predictive performance across all models.
- *Call duration* and *previous campaign outcome (poutcome)* are consistently the strongest predictors of customer subscription behavior.
- Boosting-based models more effectively capture non-linear interactions between features, including interactions involving macroeconomic indicators.
- Although macroeconomic variables have limited individual importance, their interaction effects contribute meaningfully to overall model performance.
- The alignment of evaluation metrics, confusion matrix results, ROC curves, and feature importance patterns demonstrates strong internal consistency and supports the robustness of the modeling pipeline.

4.5. Challenges and Limitations

• Challenges

The modeling approach in the notebook is subject to several notable limitations. A primary challenge arises from the imbalanced class distribution, in which subscribing customers constitute a minority class, making accurate identification of positive cases difficult and resulting in a persistent presence of false negative predictions across models. Additionally, model performance is strongly influenced by interaction-based features, particularly call duration. While this variable exhibits high predictive power, it is only available after customer contact has occurred, which limits its practical applicability in pre-contact prediction and targeting scenarios. These factors constrain the generalizability and real-world usefulness of the modeling results

• Model Limitations

From a modeling perspective, the study is subject to several limitations. Feature importance analyses are inherently model-specific and reflect relative contributions rather than causal relationships, and their

interpretation may be affected by correlations among input features. In addition, model evaluation relies on a single train–test split, which may not fully capture performance variability or ensure robustness across different data partitions. These limitations highlight constraints in the experimental design and suggest that the reported results should be interpreted with appropriate caution.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1. Conclusion

This study developed a complete machine-learning pipeline to predict customer subscription outcomes using a standardized marketing dataset. The notebook included all essential preprocessing steps, which is from data inspection and missing-value handling to categorical encoding, feature scaling, and train–test splitting, ensuring a consistent and reproducible workflow. Four main models were trained and evaluated: Random Forest, XGBoost, Gradient Boosting, and LightGBM. The comparison revealed clear performance differences among them, with the boosting-based approaches showing stronger predictive capability overall. Among the four, LightGBM achieved the highest and most stable Accuracy, F1-score, and ROC-AUC. Overfitting analysis further confirmed its strong generalization with only minimal train–test discrepancies. Overall, the results demonstrate that gradient-boosting models are well suited for capturing the complex and nonlinear structure of the marketing dataset, leading to more reliable predictions..

5.2. Future Work

Although the current analysis and models provide meaningful insights, several directions can further improve predictive performance and support better decision-making for future marketing campaigns:

- **Expand and Enrich the Dataset**

Collect additional behavioral features such as transaction frequency, savings history, and product usage level. Incorporate macroeconomic indicators (interest rates, CPI, GDP growth) to capture market context and time-dependent effects.

- **Address Class Imbalance**

Apply techniques such as SMOTE, ADASYN, or Balanced Bagging to improve the model’s ability to predict the minority “yes” class, which is significantly underrepresented.

- **Explore More Advanced Models**

Experiment with powerful algorithms such as XGBoost, LightGBM, and CatBoost for performance comparison. Use hyperparameter optimization methods (GridSearchCV, RandomizedSearch, Bayesian Optimization) to enhance model accuracy.

- **Advanced Feature Engineering**

Create new features (e.g., recency of last contact, employment stability, age grouping, customer clustering). Investigate non-linear interactions between variables to capture more complex patterns.

- **Customer Segmentation**

Combine clustering techniques with predictive modeling to design personalized marketing strategies for different customer segments.

- **Enhanced Customer Behavior Analysis**

Apply Explainable AI methods (SHAP, LIME) to better understand the impact of each feature on the model's predictions. Analyze behavioral shifts across campaigns or demographic groups.

5.3. Recommendations for Bank

Although the current analysis provides valuable insights into customer behavior and campaign performance, several areas can be further developed to enhance model accuracy and support more effective marketing strategies. Additionally, the bank can apply these findings to improve customer targeting and operational efficiency.

- **Optimize Targeting Based on Customer Profile**

Customers with higher education (university, professional courses) and older or retired clients show higher subscription rates, these groups should be prioritized. Some job categories (admin, technician, blue-collar) have large populations but low “yes” rates; marketing resources can be reallocated more efficiently.

- **Improve Contact Strategy**

Reduce repeated contacts for customers who have consistently shown low interest, this lowers campaign costs. Focus on optimal call duration and timing, as longer call durations correlate strongly with higher subscription probability.

- **Personalize Marketing Messages**

Create targeted content for each segment (e.g., retirement savings products for older groups, flexible investment plans for working professionals). Tailor communication channels: email/SMS for tech-savvy clients, phone calls for older clients.

- **Strengthen Data Quality and Customer Records**

Reduce the number of "unknown" or missing values in fields like job, education, and marital status. Implement better data-collection processes during customer onboarding and interactions.

- **Integrate Real-Time Customer Behavior Tracking**

Track engagement in digital banking apps to identify potential leads more accurately. Use behavioral scoring models to update customer likelihood dynamically.

- **Continuous Campaign Evaluation**

Compare results across campaigns to learn what strategies work best. Use A/B testing to measure the effectiveness of different scripts, offers, or contact schedules.

REFERENCES

- [1] Cortez, P., & Moro, S. (2010). Bank marketing dataset. UCI Machine Learning Repository.
- [2] Moro, S., Cortez, P., & Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. *Decision Support Systems*, 62, 22–31.
- [3] Almeida, M., Moro, S., & Laureano, R. (2012). Bank marketing: Leveraging the data for optimized campaign management. University of Minho.
- [4] Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5–32.
- [5] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.
- [6] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- [7] Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3), 379–423.